

Project charter: Connection Optimization and Evaluation for Next-Generation Curtainwall Energy Technology – iWin

General information

Project acronym	CONNECT-iWin
Project manager	Denys Kovalov
Sponsor	Rubén Roldan Molinero
Organizational unit	N.A.
Expected start date	01.03.2026
Expected completion date	31.05.2026

Background

The purpose of this project is to identify, evaluate, and select the most appropriate electrical connection system for building-integrated photovoltaics (BIPV) in the form of Active Windows.

The increasing adoption of glass-dominant façades and distributed energy generation in buildings requires electrical connection solutions that ensure safety, scalability, cost efficiency, architectural compatibility, and optimal energy performance.

A structured comparative evaluation tailored specifically to photovoltaic windows has not yet been conducted. This project aims to reduce technical, regulatory, and commercial risks prior to large-scale industrialization.

Goals

Key ID	Specific	Metric	Achievable	Relevance	Time
1	Develop and document a comprehensive list of feasible electrical connection architectures for Active Windows considering string systems, microinverters, power optimizers with central inverter, DC bus per floor, hybrid AC/DC systems.	Minimum of 3 architectures formally documented, including topology diagrams, operating ranges, and benchmark references.	Executed by DK using market research, technical benchmarking and expertise.	Reduces risk of suboptimal technology selection.	M1
2	Perform structured technical, regulatory, and financial evaluation of shortlisted architectures.	Evaluation of shortlisted options through a weighted Multi-Criteria Decision Analysis (MCDA)	Based on a previously-defined evaluation framework.	Enables objective, data-driven decision aligned with risk management and cost control processes.	M2
3	Definition of a baseline for Technical Requirements Specification (TRS) for the selected architecture.	TRS document issued including voltage levels, protection schemes, connector standards, wiring topology, BMS integration, and monitoring architecture.	Derived from evaluation results and aligned with IEC/NEC standards	Baseline for subsequent design and development phases	M3

Scope

Included in Scope:

- State-of-the-art review in BIPV electrical architectures.
- Technical benchmarking of existing solutions.
- Regulatory analysis (IEC, NEC, local building codes).
- Preliminary electrical modeling and loss estimation.
- Thermal and safety considerations.
- Economic comparative analysis.
- Final technical recommendation.

Out of Scope:

- Detailed product engineering design.
- Certification processes.
- Pilot manufacturing.
- Structural façade engineering.

Key stakeholders

Name	Project Role	Organization	Contact information
Denys Kovalov	Project manager & technical expert	N.A	N.A
Ruben Roldan Molinero	Sponsor and technical support	N.A	N.A

Project Milestones

Key Milestone	Deadline	Notes
M1 - Architecture shortlist approved	31.03.2026	D1 – Electrical connection architectures including topology diagrams, operating ranges and benchmark references
M2 - Evaluation dataset validated	30.04.2026	D2 - weighted Multi-Criteria Decision Analysis (MCDA)
M3 - Technical Requirements Specification approved	31.05.2026	D3 - Technical Requirements Specification document

Constrains, Assumptions, risks and dependencies

Risk [high level perspective]	Analysis and assessment
R1 - Electrical safety risk due to high-voltage DC integration in façade elements.	<u>Probability</u> : 4 <u>Impact</u> : 5 <u>Preventive/Corrective measure</u> : Limit system voltage; adopt module-level power electronics; integrate DC arc-fault detection; early compliance review (IEC/NEC); third-party safety audit. <u>Risk score</u> : 8/25
R2 - Regulatory non-compliance or delayed certification (BIPV + façade integration)	<u>Probability</u> : 3 <u>Impact</u> : 5 <u>Preventive/Corrective measure</u> : Early engagement with certification bodies; regulatory mapping; design-to-standard approach; pre-compliance testing. <u>Risk score</u> : 6/25
R3 - Excessive façade wiring complexity increasing installation time and cost.	<u>Probability</u> : 4 <u>Impact</u> : 4 <u>Preventive/Corrective measure</u> : Optimize topology (bus-per-floor or modular design); installation simulation; involve EPC early; prefabricated harness strategy. <u>Risk score</u> : 6/25
R4 - CAPEX higher than market-acceptable threshold.	<u>Probability</u> : 4 <u>Impact</u> : 4 <u>Preventive/Corrective measure</u> : Detailed cost modeling; supplier pre-negotiations; design-to-cost workshops; modular architecture to reduce electronics count. <u>Risk score</u> : 9/25

R5 - Reliability issues due to distributed electronics in façade environment (temperature, humidity).	<u>Probability</u>: 3 <u>Impact</u>: 5 <u>Preventive/Corrective measure</u>: Environmental stress testing; IP-rated enclosures; reduce component count; redundancy strategy. <u>Risk score</u>: 6/25
R6 - Retrofit incompatibility with existing building electrical infrastructure.	<u>Probability</u>: 3 <u>Impact</u>: 4 <u>Preventive/Corrective measure</u>: Develop hybrid AC/DC options; standardized interface box; retrofit case studies; early pilot building validation. <u>Risk score</u>: 4/25
R7 - BMS integration complexity or interoperability failure.	<u>Probability</u>: 3 <u>Impact</u>: 3 <u>Preventive/Corrective measure</u>: Use standard protocols (KNX/Modbus/BACnet); early interface specification; integration testing phase. <u>Risk score</u>: 4/25
R8 - Technology selection bias leading to suboptimal architecture decision.	<u>Probability</u>: 2 <u>Impact</u>: 4 <u>Preventive/Corrective measure</u>: Formal Multi-Criteria Decision Analysis (MCDA); sensitivity analysis $\pm 20\%$; independent technical review. <u>Risk score</u>: 3/25
R9 - Supply chain dependency on specialized power electronics.	<u>Probability</u>: 3 <u>Impact</u>: 4 <u>Preventive/Corrective measure</u>: Dual supplier strategy; early supplier qualification; long-lead component mapping. <u>Risk score</u>: 4/25
R10 - Investor hesitation due to perceived technical risk.	<u>Probability</u>: 3 <u>Impact</u>: 5 <u>Preventive/Corrective measure</u>: Clear quantitative risk mitigation plan; pilot demonstration; staged development roadmap. <u>Risk score</u>: 6/25
R11 - Project timeline overrun due to iterative evaluation cycles.	<u>Probability</u>: 3 <u>Impact</u>: 3 <u>Preventive/Corrective measure</u>: Defined phase gates; strict evaluation criteria; weekly technical review meetings; contingency buffer. <u>Risk score</u>: 4/25

Project authorization

By signing below, I _____, the Project Sponsor, approve/decline the project described herein and authorize/deny it to begin.

iWin – innovative Windows SA

Authorized signature of the CEO

Signature of the project sponsor